

## MENADO SAM RATULANGI, Indonesia

WMO: 970140

Lat: 1.5458N

Lon: 124.9230E

Elev: 262

StdP: 14.56

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | 69.4       | 70.7 | 60.8                          | 80.2 | 82.4 | 63.1 | 87.1 | 81.5 | 14.0                  | 84.7 | 12.7 | 85.1 | 1.8                   | 30   | 0.279 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 8             | 16.8                   | 93.1            | 76.4 | 91.6 | 76.4 | 90.8 | 76.5 | 80.0                | 87.9 | 79.5 | 87.1 | 79.1 | 86.5 | 7.8                  | 180  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 77.8                            | 145.9 | 83.5 | 77.3 | 143.5 | 82.9 | 77.1 | 142.3 | 82.6 | 43.9          | 88.5 | 43.2 | 87.6 | 42.7 | 87.0 | 86.5              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |       |     |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|-------|-----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |       |     |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |       |     |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |       |     |
| (4) 16.6          | 13.5 | 11.1 | DB                         | 66.1 | 94.8               | 3.4 | 2.5                                                | 63.6 | 96.6       | 61.6 | 98.1       | 59.7 | 99.5       | 57.2 | 101.3 | (4) |
| (5)               |      |      | WB                         | 62.4 | 82.9               | 3.3 | 2.0                                                | 60.1 | 84.3       | 58.2 | 85.5       | 56.3 | 86.6       | 54.0 | 88.1  | (5) |

## Monthly Climatic Design Conditions

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 80.6   | 79.6  | 79.4  | 80.2  | 80.7  | 81.3  | 80.8  | 80.8  | 81.3  | 81.2  | 81.1  | 80.6  | 80.3 |
|                                                                               | DBStd       | 1.75   | 1.64  | 1.72  | 1.54  | 1.52  | 1.55  | 1.76  | 1.76  | 1.91  | 1.65  | 1.56  | 1.55  | 1.64 |
|                                                                               | HDD50       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                               | HDD65       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                               | CDD50       | 11174  | 918   | 824   | 936   | 921   | 969   | 924   | 955   | 970   | 935   | 965   | 918   | 939  |
|                                                                               | CDD65       | 5699   | 453   | 404   | 471   | 471   | 504   | 474   | 490   | 505   | 485   | 500   | 468   | 474  |
|                                                                               | CDH74       | 51457  | 3605  | 3172  | 4052  | 4207  | 4711  | 4328  | 4676  | 5126  | 4830  | 4672  | 4027  | 4051 |
|                                                                               | CDH80       | 18573  | 971   | 898   | 1327  | 1448  | 1791  | 1608  | 1787  | 2094  | 2097  | 1912  | 1389  | 1251 |
| Wind                                                                          | WSAvg       | 3.7    | 3.1   | 3.4   | 3.5   | 2.9   | 2.8   | 3.5   | 5.1   | 6.1   | 4.5   | 3.4   | 2.7   | 3.1  |
| Precipitation                                                                 | PrecAvg     | 123.7  | 18.1  | 14.7  | 13.1  | 10.8  | 10.0  | 8.4   | 6.1   | 5.0   | 6.9   | 7.8   | 12.8  | 14.2 |
|                                                                               | PrecMax     | 173.5  | 39.5  | 31.9  | 30.4  | 19.8  | 18.5  | 16.5  | 15.7  | 14.9  | 21.7  | 24.2  | 24.8  | 23.1 |
|                                                                               | PrecMin     | 76.3   | 2.1   | 1.5   | 2.5   | 4.6   | 3.0   | 0.2   | 0.0   | 0.0   | 0.0   | 0.1   | 2.6   | 3.0  |
|                                                                               | PrecStd     | 24.3   | 9.1   | 7.1   | 7.4   | 4.1   | 3.5   | 4.2   | 4.1   | 3.7   | 6.0   | 6.2   | 6.2   | 5.7  |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 89.8  | 89.8  | 91.3  | 91.6  | 93.1  | 92.4  | 92.8  | 93.6  | 95.0  | 94.7  | 91.8  | 91.3 |
|                                                                               |             | MCWB   | 78.2  | 77.9  | 77.9  | 78.7  | 79.6  | 78.3  | 77.6  | 76.0  | 74.5  | 75.2  | 77.6  | 78.8 |
|                                                                               | 2%          | DB     | 87.8  | 88.0  | 89.2  | 89.7  | 91.0  | 90.7  | 90.7  | 91.6  | 92.7  | 91.8  | 89.9  | 89.0 |
|                                                                               |             | MCWB   | 77.8  | 77.5  | 77.7  | 78.2  | 78.3  | 77.2  | 76.2  | 75.1  | 74.6  | 76.1  | 77.5  | 78.3 |
|                                                                               | 5%          | DB     | 86.3  | 86.5  | 87.7  | 88.2  | 89.7  | 89.4  | 89.4  | 90.2  | 91.1  | 90.2  | 88.3  | 87.5 |
|                                                                               |             | MCWB   | 77.6  | 77.2  | 77.5  | 78.0  | 78.1  | 76.9  | 75.9  | 75.0  | 75.0  | 76.5  | 77.7  | 78.2 |
|                                                                               | 10%         | DB     | 85.0  | 85.4  | 86.3  | 87.1  | 88.1  | 87.9  | 88.0  | 89.1  | 89.6  | 88.6  | 87.0  | 86.0 |
|                                                                               |             | MCWB   | 77.2  | 77.0  | 77.3  | 77.8  | 78.0  | 76.9  | 75.8  | 75.0  | 75.6  | 76.7  | 77.7  | 77.9 |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 79.7  | 79.3  | 79.8  | 80.3  | 80.9  | 80.1  | 79.5  | 79.1  | 79.2  | 79.8  | 80.0  | 80.1 |
|                                                                               |             | MCDB   | 86.5  | 86.4  | 87.3  | 88.1  | 89.1  | 88.4  | 88.3  | 88.6  | 88.3  | 88.4  | 87.5  | 87.3 |
|                                                                               | 2%          | WB     | 78.8  | 78.6  | 78.9  | 79.3  | 79.9  | 79.1  | 78.4  | 77.9  | 78.4  | 78.9  | 79.1  | 79.2 |
|                                                                               |             | MCDB   | 85.2  | 85.6  | 86.2  | 86.8  | 87.7  | 87.0  | 86.8  | 87.4  | 87.3  | 87.4  | 86.4  | 85.9 |
|                                                                               | 5%          | WB     | 78.2  | 78.0  | 78.3  | 78.8  | 79.1  | 78.3  | 77.6  | 77.1  | 77.7  | 78.2  | 78.5  | 78.8 |
|                                                                               |             | MCDB   | 84.2  | 84.5  | 85.2  | 86.1  | 86.6  | 85.8  | 85.8  | 86.4  | 86.5  | 86.4  | 85.6  | 85.2 |
|                                                                               | 10%         | WB     | 77.6  | 77.3  | 77.8  | 78.2  | 78.5  | 77.7  | 76.9  | 76.3  | 76.9  | 77.6  | 78.0  | 78.2 |
|                                                                               |             | MCDB   | 83.3  | 83.4  | 84.4  | 85.0  | 85.7  | 85.0  | 84.9  | 85.3  | 85.5  | 85.5  | 84.8  | 84.2 |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 11.6  | 12.0  | 13.4  | 13.9  | 15.3  | 15.3  | 15.6  | 16.8  | 18.4  | 17.1  | 14.3  | 12.6 |
|                                                                               |             | MCDBR  | 13.8  | 14.5  | 15.7  | 15.8  | 17.3  | 17.4  | 17.7  | 18.8  | 20.9  | 19.5  | 16.4  | 14.8 |
|                                                                               |             | MCWBR  | 5.6   | 6.0   | 6.4   | 6.6   | 7.0   | 7.0   | 7.1   | 7.4   | 7.9   | 7.4   | 6.6   | 6.1  |
|                                                                               | 5% WB       | MCDBR  | 12.7  | 13.2  | 14.1  | 14.6  | 15.7  | 15.6  | 15.9  | 17.1  | 17.6  | 16.6  | 14.9  | 13.6 |
|                                                                               |             | MCWBR  | 5.6   | 5.9   | 6.3   | 6.6   | 6.8   | 6.7   | 6.8   | 7.1   | 7.3   | 6.9   | 6.5   | 6.1  |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.417  | 0.422 | 0.425 | 0.426 | 0.420 | 0.418 | 0.414 | 0.424 | 0.437 | 0.477 | 0.440 | 0.422 |      |
|                                                                               | taud        | 2.486  | 2.462 | 2.450 | 2.441 | 2.464 | 2.483 | 2.497 | 2.454 | 2.402 | 2.276 | 2.394 | 2.464 |      |
|                                                                               | Ebn at Noon | 289    | 289   | 286   | 279   | 274   | 271   | 273   | 276   | 278   | 269   | 280   | 285   |      |
|                                                                               | Edh at Noon | 36     | 37    | 38    | 37    | 35    | 34    | 34    | 36    | 39    | 44    | 39    | 36    |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 1546   | 1681  | 1793  | 1798  | 1741  | 1614  | 1711  | 1850  | 1943  | 1869  | 1715  | 1604  |      |
|                                                                               | RadStd      | 98     | 142   | 149   | 128   | 106   | 102   | 150   | 108   | 121   | 107   | 104   | 118   |      |

## Historical Trends

|                            | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|----------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|                            |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                            |       | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)   | (k)   | (l)   |
| (46) Station Only          | +0.54 | N/A     | N/A    | +1.69   | +1.13 | +0.81 | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (1 neighbor) | +0.54 | N/A     | N/A    | +1.69   | +1.13 | +0.81 | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page